Complex Rays and the Hyperbolic Plane:
why the ABCD law $q' = (Aq + B)/(Cq + D)$ holds

An expository note

October 2026

Abstract

A ray-transfer (ABCD) matrix also transforms the complex beam parameter q of a Gaussian beam, by a Möbius map. This note explains why. In one sentence: *a Gaussian beam is a complex ray.* The idea comes from Kogelnik, Deschamps and Arnaud. Treating paraxial optics as quantum mechanics shows the idea is exact, not an analogy. Three things follow. The ABCD law holds because q is the projective coordinate of that ray, which is what makes the map Möbius. The same complex ray also gives the beam's magnification and Gouy phase. And Gaussian beams form the hyperbolic plane, on which optical systems act as isometries. Mode matching, resonator stability and the Gouy phase all then have direct geometric meanings. We summarize the literature along the way and note the few points where we found no prior statement.

1 The law, and what needs explaining

A paraxial ray at height x and small angle θ passes through a lossless optical system as

$$\begin{pmatrix} x' \\ \theta' \end{pmatrix} = M \begin{pmatrix} x \\ \theta \end{pmatrix}, \qquad M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}; \qquad \underbrace{\begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}}_{\text{free space}}, \quad \underbrace{\begin{pmatrix} 1 & 0 \\ -1/f & 1 \end{pmatrix}}_{\text{thin lens}}. \tag{1}$$

A Gaussian beam in one transverse dimension (two dimensions come in Section 8) is

$$u_q(x) = \exp\Big(\frac{ikx^2}{2q}\Big), \qquad \frac{1}{q} = \frac{1}{R} + \frac{i\lambda}{\pi w^2}, \tag{2}$$

where R is the wavefront radius of curvature and w the spot size. Kogelnik stated and named the *ABCD law* in 1965 [1] (see also [2, 3]): the output is again Gaussian, with

$$\boxed{q' = \frac{Aq + B}{Cq + D}} \tag{3}$$

The usual argument and its gaps. One checks (3) for free space ($q \mapsto q + d$) and for a thin lens ($1/q \mapsto 1/q - 1/f$). Then one decomposes an arbitrary system into such elements, using the fact that Möbius maps compose like their matrices. Kogelnik's original proof is complete for every system with $C \neq 0$: it reduces such a system to free space, a lens and free space, using principal planes [1]. Erden and Ozaktas give the equivalent three-factor decomposition, valid for $C \neq 0$ [4]. Two factors do not suffice in general, and afocal systems ($C = 0$) need a separate

step. The real problem, though, is that the argument *verifies* the law without explaining it. Why should a matrix that describes rays describe the exact diffraction of a wave? Why a Möbius map? What happens to the overall amplitude and phase? Kogelnik and Li gave the right hint [3]:

> "The ABCD law follows from the analogy between the laws for laser beams and the laws obeyed by the spherical waves in geometrical optics."

This note explains why that analogy is exact, and what follows from it.

Conventions. We use the physics convention $\mathrm{Re}[u\,e^{i(kz-\omega t)}]$. Then a beam at distance z past a waist of Rayleigh range z_R has $q = z - iz_R$, so physical beams have $\mathrm{Im}\,q < 0$. Kogelnik, Siegman and most laser texts use $e^{+i\omega t}$. That convention complex-conjugates everything and gives $q = z + iz_R$. Real ABCD matrices commute with complex conjugation, so every result below holds in either convention. In figures we plot $\tau = \bar{q} = z + iz_R$ in the upper half-plane $\mathbb{H}$.

2 Rays: the symplectic group

With the reduced angle $n\,dx/dz$ as momentum, paraxial ray optics is Hamiltonian mechanics with z playing the role of time and a Hamiltonian quadratic in (x, θ). Its flow is therefore linear, $r \mapsto Mr$ with $r = (x, \theta)^T$, and it preserves the symplectic form

$$\omega(r_1, r_2) = x_1\theta_2 - \theta_1 x_2. \tag{4}$$

In one transverse dimension, preserving ω just means $\det M = AD - BC = 1$. So the ray matrices form $\mathrm{SL}(2,\mathbb{R}) = \mathrm{Sp}(2,\mathbb{R})$; with two transverse dimensions they form $\mathrm{Sp}(4,\mathbb{R})$ [5–7]. Physically, this is Liouville's theorem, or conservation of étendue. Kogelnik derived $AD - BC = 1$ from reciprocity [1]. All the structure below rests on this one condition.

3 The central idea: a Gaussian beam is a complex ray

Spherical waves. Consider a point source a distance R behind a reference plane. Every ray from it satisfies $x = R\theta$, so $R = x/\theta$. Apply (1) to any such ray:

$$R' = \frac{x'}{\theta'} = \frac{Ax + B\theta}{Cx + D\theta} = \frac{AR + B}{CR + D}.$$

This is the geometrical-optics imaging law for spherical waves: the wavefront radius obeys the ABCD law. Paraxially, a spherical wave is $\exp(ikx^2/2R)$, which is (2) with q real. This is the analogy Kogelnik and Li point to.

Complexify. The ray equations are linear with real coefficients, so complex solutions propagate by the same matrix. A Gaussian beam is a "spherical wave" whose radius is complex. Equivalently, it is the field of a point source at a complex position [8]: $q = z - iz_R$ is literally the complex distance to a source at $z = iz_R$. Deschamps [8] and Keller and Streifer [9] made this precise as complex rays in 1971. Arnaud worked it out most fully. In his *complex ray pencils* [10, 11], the wavefront curvature is the ratio of a ray's angle to its height, and the field amplitude comes

from the ray-tube area. He remarks that "it has nowhere been specified that [the rays] are real quantities." So the Gaussian beam with parameter q corresponds to the complex ray

$$v = \begin{pmatrix} x \\ \theta \end{pmatrix} = \begin{pmatrix} q \\ 1 \end{pmatrix} \in \mathbb{C}^2, \qquad q = \frac{x}{\theta}, \tag{5}$$

defined up to an overall complex factor. Nearly everything about the beam can be read off this one vector.

1. **The ABCD law** is the ratio: $Mv = (Aq + B,\ Cq + D)^T \propto (q', 1)^T$. The map is linear-fractional because q is a *projective* coordinate. The beam is a complex line $[v] \in \mathbb{C}P^1$, and a linear map of $\mathbb{C}^2$ always acts on lines by a Möbius map.

2. **Magnification and Gouy phase** come from the change in the ray's height:

$$\frac{x'}{x} = A + \frac{B}{q} = \frac{w'}{w}\, e^{i\psi}, \tag{6}$$

   where ψ is the Gouy phase accumulated through the system (Section 5). The modulus identity follows from (2) in two lines. The phase identity is the Erden–Ozaktas formula [4], rewritten. In Arnaud's language, the amplitude goes as $|x|^{-1/2}$ (energy conservation in a ray tube) and the phase is that of the complex tube width.

3. **Confinement.** For v and its conjugate $\bar{v}$, the Lagrange invariant is $\omega(\bar{v}, v) = \bar{q} - q = 2i\,|\operatorname{Im} q|$ (for v normalized as in (5)). It is conserved because M is real and symplectic, so the sign of $\operatorname{Im} q$, and with it normalizability, can never change.

4. **Resonator modes** are complex rays that are eigenvectors of the round-trip matrix (Section 7). Vlasov and Talanov already noted (as quoted by Arnaud [10]) that the phase a matched beam picks up equals the phase of a ray-matrix eigenvalue.

The complex-ray picture explains why the law has the form it does. It does not yet justify itself: why should a wave obey the equations of complexified rays, exactly and including diffraction? That is the job of the next section.

4 Why complex rays are exact: paraxial optics is quantum mechanics

The paraxial equation $2ik\,\partial_z u + \partial_x^2 u = 0$, divided by $2k$ with $\lambda\!\!\!^{-} = 1/k = \lambda/2\pi$, reads

$$i\lambda\!\!\!^{-}\,\partial_z u = \tfrac{1}{2}\hat{p}^{\,2} u, \qquad \hat{p} = -i\lambda\!\!\!^{-}\,\partial_x, \qquad [\hat{x}, \hat{p}] = i\lambda\!\!\!^{-}. \tag{7}$$

This is the Schrödinger equation for a free particle, with $\hbar \to \lambda\!\!\!^{-}$, time $\to z$, and momentum $\to$ ray angle. Gloge and Marcuse first set out this dictionary [12]. Stoler and Nazarathy–Shamir developed the operator methods [13, 14]. A thin lens is the impulsive potential $\hat{x}^2/2f$, and a graded-index medium is a harmonic oscillator. Every first-order element therefore has a Hamiltonian at most quadratic in $(\hat{x}, \hat{p})$.

Heisenberg evolution is ray optics. For a quadratic Hamiltonian, the Heisenberg equations for $\hat{x}, \hat{p}$ are linear and *identical* to the ray equations. For free space, for example, $d\hat{x}/dz = \hat{p}$ and $d\hat{p}/dz = 0$. So if U is the unitary wave propagator of a system with ray matrix M,

$$\boxed{U^\dagger \hat{r}\, U = M\hat{r}, \qquad \hat{r} = (\hat{x}, \hat{p})^T} \tag{8}$$

exactly. Ehrenfest's theorem has no corrections here: the centroid of any beam follows a ray.

The Gaussian as an annihilated state. Since $\hat{p}\, u_q = (x/q)\, u_q$, we have

$$(q\hat{p} - \hat{x})\, u_q = 0, \qquad \text{i.e.} \qquad \omega(v, \hat{r})\, u_q = 0 \;\; \text{with } v = (q, 1)^T. \tag{9}$$

This first-order ODE determines u_q up to a constant. A Gaussian beam is thus *the state annihilated by the phase-space operator of its complex ray.* In quantum-optics terms it is a squeezed vacuum: $\omega(v, \hat{r})$ and $\omega(\bar{v}, \hat{r})$ are rescaled annihilation and creation operators, since their commutator $i\lambda\, \omega(v, \bar{v}) = -2\lambda \operatorname{Im} q$ is positive exactly when the beam is normalizable.

Theorem 1 (ABCD law, operator form)**.** *If $\omega(v, \hat{r})\, u = 0$, then $\omega(Mv, \hat{r})\, (Uu) = 0$. Hence $Uu_q \propto u_{q'}$ with $q' = (Aq + B)/(Cq + D)$, for every $M \in \mathrm{SL}(2, \mathbb{R})$, including $C = 0$.*

Proof. By (8), $U^\dagger \omega(Mv, \hat{r}) U = \omega(Mv, M\hat{r}) = \omega(v, \hat{r})$, since M is symplectic. So $\omega(Mv, \hat{r})\, Uu = U\omega(v, \hat{r})u = 0$, and $Mv \propto (q', 1)^T$. □

This is the complex-ray statement of Section 3, now exact at the level of waves. The same argument appears in the semiclassical wave-packet literature. Littlejohn applies it to the standard Gaussian (Eq. 6.32 of [15]). Arvind, Dutta, Mukunda and Simon give it for $\mathrm{Sp}(2n, \mathbb{R})$ [16]. Simon, Sudarshan and Mukunda state Kogelnik's law as the $n = 1$ case of a matrix Möbius action on Gaussian states [17].

Two other derivations, and what they show. (i) *Diffraction integral.* The propagator of any first-order system is the Collins integral [18, 19]

$$(Uu)(x') = \frac{1}{\sqrt{i\lambda B}} \int \exp\Big[\frac{ik}{2B}\big(Ax^2 - 2xx' + Dx'^2\big)\Big] u(x)\, dx. \tag{10}$$

Its exponent is k times the optical path along the unique ray from x to x' (Hamilton's point characteristic). Feeding in u_q is a Gaussian integral; completing the square gives (3) together with the prefactor $(A + B/q)^{-1/2}$. This is the most direct computation [4]. It shows that the wave phase is the ray path length, a statement of stationary phase that is exact for quadratic exponents. (ii) *Riccati linearization.* Put a Gaussian ansatz into the paraxial equation for a continuously varying lenslike medium. You get a Riccati equation for $1/q$. Writing $1/q = \theta/x$ turns it into the linear ray equation for (x, θ). This is the standard trick for Riccati equations, and in the wave-packet literature it is exactly Heller's and Hagedorn's parametrization $\mathcal{C} = PQ^{-1}$ by a "complex ray" (Q, P) [20–22].

All three derivations say the same thing. The complex ray is the fundamental object, and the Möbius map is its shadow on $\mathbb{C}P^1$.

5 The leftover scalar: amplitude, Gouy phase, and a sign

Theorem 1 gives $Uu_q = c\,u_{q'}$. With $u_q(0) = 1$, the Collins integral gives

$$Uu_q = \left(A + \frac{B}{q}\right)^{-1/2} u_{q'} \qquad \text{(per transverse dimension)}. \tag{11}$$

So the prefactor is $(x'/x)^{-1/2}$ for the complex ray. Its modulus $(w/w')^{1/2}$ conserves power, and its phase is $-\frac{1}{2}\psi$ with $\psi = \arg(A + B/q)$. Two transverse dimensions square the prefactor. Free space from a waist gives $A + B/q = 1 + iz/z_R$ and $\psi = \arctan(z/z_R)$, the textbook Gouy phase. For a general system, $\psi = \arg(A + B/q_{\text{in}})$ is the *accumulated* Gouy phase of Erden and Ozaktas [4]. The Hermite–Gauss mode HG_n (standard real-argument form) acquires $e^{-i(n+\frac{1}{2})\psi}$: each application of the creation operator $\omega(\bar{v}, \hat{r})$ adds one more unit of ψ. We checked this numerically for $n \le 3$ (Appendix A).

The scalar $Cq + D$ by which the ray v is rescaled ($Mv = (Cq + D)v'$) is the classical *automorphy factor* of the $\text{SL}(2,\mathbb{R})$ action on the half-plane. It obeys the chain rule $j(M_2M_1, q) = j(M_2, M_1q)\, j(M_1, q)$, and $A + B/q = j(M, q)\, q'/q$. This is why Gouy phases add along a cascaded system.

The sign, and the metaplectic group. The square root in (11) is genuinely ambiguous. Take a 1D graded-index guide, a harmonic oscillator. After one full ray period $M = I$, yet the Gaussian mode returns as $-u$: it has picked up the oscillator's zero-point phase $e^{-i\pi}$ (we confirm this numerically). The ray matrix therefore fixes the wave operator only up to sign. The operators $\pm U(M)$ form the *metaplectic group* $\text{Mp}(2,\mathbb{R})$, the double cover of $\text{SL}(2,\mathbb{R})$. Bacry and Cadilhac identified it as the group of paraxial wave optics, with ray optics its image in $\text{SL}(2,\mathbb{R})$ [23]. Littlejohn [15] and Arvind et al. [16] discuss the sign and its relation to the Maslov index. Topologically, $\pi_1(\text{SL}(2,\mathbb{R})) = \mathbb{Z}$, and the phase depends on the homotopy class of the path of systems [24]. The practical upshot [25]: accumulate ψ continuously, $\psi = \sum_i \arg(1 + d_i/q_i)$ over the free-space sections, rather than taking the principal value of $\arg(A + B/q)$. In our numerical test system the two differ by 2π, so the principal value gives the wrong overall sign for a 1D field. For a circular beam in two dimensions the prefactor is $(A + B/q)^{-1}$ and the sign cancels. Even then, the accumulated real-valued Gouy phase, not its value mod 2π, is what sets the transverse-mode spacing of a resonator.

6 Phase space: Wigner functions, beam matrices, and M^2

There is a second route, one level down, that forgets the phase. For quadratic Hamiltonians the Wigner function is transported rigidly along rays, $W'(r) = W(M^{-1}r)$ [26]. So the second-moment ("beam") matrix of *any* beam, coherent or not,

$$\Sigma = \begin{pmatrix} \langle x^2 \rangle & \langle x\theta \rangle \\ \langle x\theta \rangle & \langle \theta^2 \rangle \end{pmatrix} \longmapsto M\Sigma M^T, \tag{12}$$

transforms exactly like the covariance of a bundle of rays. Bélanger showed that this linear law, together with the invariant $\det \Sigma$, is *equivalent* to a Möbius law for a complex Q [27]. That gives an ABCD law for arbitrary beams, with the invariant M^2. For a Gaussian beam,

$$\Sigma = \frac{\lambda}{2|\operatorname{Im} q|} \operatorname{Re}\left(vv^\dagger\right) = \frac{\lambda}{2|\operatorname{Im} q|} \begin{pmatrix} |q|^2 & \operatorname{Re} q \\ \operatorname{Re} q & 1 \end{pmatrix}, \qquad \det \Sigma = \left(\frac{\lambda}{2}\right)^2, \tag{13}$$

which is the complex ray again, now as a real outer product. A geometric reading follows. Symmetric 2×2 matrices with the quadratic form det make a Minkowski space $\mathbb{R}^{2,1}$, and $\Sigma \mapsto M\Sigma M^T$ is a Lorentz transformation, so $\mathrm{SL}(2,\mathbb{R})$ acts as $\mathrm{SO}^+(2,1)$. The pure Gaussians fill the "mass shell" $\det \Sigma = (\lambda/2)^2$, a hyperboloid. The invariant *mass* $\sqrt{\det \Sigma}/(\lambda/2)$ is the beam-quality factor M^2. Gaussian beams sit at $M^2 = 1$, where they saturate the uncertainty principle. Other beams lie on heavier shells, and no lossless first-order system can change M^2. Ohsawa and Tronci state the multidimensional version of (13) [28].

This route is intuitive and covers partially coherent light [7]. It cannot see the Gouy phase or the metaplectic sign, which is why the wave-level argument of Section 4 is still needed.

7 Geometry: the space of Gaussian beams is the hyperbolic plane

The homogeneous space. $\mathrm{SL}(2,\mathbb{R})$ acts transitively on beams: any beam is a waist of some size propagated some distance. The systems that fix a reference waist (up to phase) are the fractional Fourier transformers, i.e. sections of a lens guide matched to that beam. They form a rotation group $\mathrm{SO}(2)$. Hence

$$\{\text{Gaussian beams}\} \;\cong\; \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2) \;\cong\; \mathbb{H}, \tag{14}$$

the hyperbolic (Poincaré) half-plane. Optical systems act on it by isometries of the metric $ds^2 = |dq|^2/(\operatorname{Im} q)^2$ [16, 29, 30]. The Iwasawa decomposition, rotation × magnifier × free space or lens [30, 31], makes this concrete. Free space translates $\tau = \bar{q}$ horizontally. A lens moves it along a horocycle tangent to the real axis at 0. A magnifier dilates about 0. The boundary $\mathbb{R} \cup \{\infty\}$ consists of real rays: point sources and plane waves, which are not beams [29]. Figure 1(a) shows a path through a lens system.

Mode matching is hyperbolic distance. Propagation is unitary, so the overlap of two beams is invariant under every optical system. The only invariant of a pair of points in $\mathbb{H}$ is their hyperbolic distance d, so the overlap must be a function of d alone. The Gaussian overlap integral gives

$$\eta = \operatorname{sech}^2 \frac{d}{2} \quad \text{(circular beams)}, \qquad \tanh \frac{d}{2} = \left| \frac{q_1 - q_2}{q_1 - \bar{q}_2} \right|, \quad \cosh d = 1 + \frac{|q_1 - q_2|^2}{2 \operatorname{Im} q_1 \operatorname{Im} q_2}, \tag{15}$$

and $\operatorname{sech}(d/2)$ per transverse dimension. The pieces are in the literature but not this statement. The mode-mismatch formula used in gravitational-wave interferometry, $1 - \eta = |(q_1 - q_2)/(q_1 - \bar{q}_2)|^2$ [32, 33], is *literally* $\tanh^2(d/2)$. Rabei et al. showed that the overlap (Fubini–Study) metric on Gaussian states is $\frac{1}{8}$ of the Poincaré metric [34], which is the infinitesimal form $1 - \operatorname{sech}(d/2) \approx d^2/8$. And Gaussians are $\mathrm{SU}(1,1)$ coherent states, whose overlaps are functions of d for the same symmetry reason [35]. For two waists w_1, w_2 a distance Δz apart, (15) reproduces the familiar $\eta = 4/[(w_1/w_2 + w_2/w_1)^2 + (\lambda \Delta z/\pi w_1 w_2)^2]$.

Resonators are isometry classes. A cavity eigenmode is a fixed point of the round-trip Möbius map. Isometries of $\mathbb{H}$ are classified by trace [36, 37], and the classes are exactly the cavity types [29, 30, 38]:

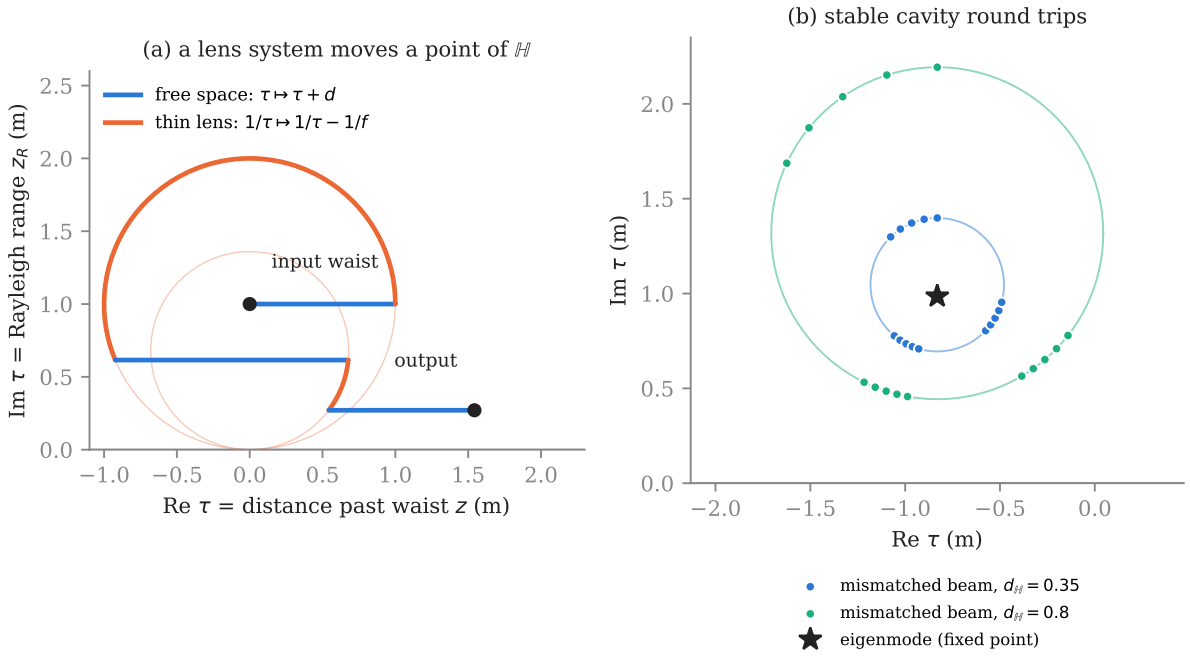


Figure 1: Gaussian beams as points $\tau = \bar{q} = z + iz_R$ of the upper half-plane. (a) A system made of free space, a lens ($f = 0.8$), free space, a lens ($f = -1.5$) and free space, starting from a waist with $z_R = 1$. Free space is a horizontal translation, and a lens moves the point along a horocycle (faint circles). (b) A stable two-mirror cavity ($R_1 = 2$, $R_2 = 3$, $L = 1.2$; lengths in m). The round trip is an elliptic isometry fixing the eigenmode ($\star$). Mismatched beams circle it on hyperbolic circles, keeping a fixed overlap with the eigenmode.

- *Elliptic*, $|A+D| < 2$: one fixed point inside $\mathbb{H}$. This is the stable cavity, with the eigenmode of Kogelnik and Li [3]. The round trip is a hyperbolic rotation about the mode, and mismatched beams circle it (Fig. 1b).
- *Parabolic*, $|A + D| = 2$: one fixed point on the boundary. This is marginal stability, e.g. plane-parallel mirrors. The symmetric confocal cavity has $M = -I$, which acts trivially: every beam is self-consistent, the familiar confocal degeneracy.
- *Hyperbolic*, $|A + D| > 2$: two fixed points on the boundary. These are real rays, the geometric-optics eigenmodes of Siegman's unstable resonators. Every beam drifts to the boundary.

The complex ray of the eigenmode is an eigenvector, $Mv_* = \mu v_*$ with $\mu = Cq_* + D = e^{i\Theta}$, $\cos\Theta = \frac{1}{2}(A+D)$. By (6) the same μ is the round-trip Gouy factor, so HG_{mn} acquires $(m+n+1)\Theta$ per round trip. For a two-mirror cavity this reproduces the standard $\Theta = 2\arccos\sqrt{g_1 g_2}$ (for $g_1, g_2 > 0$) [3]. Since $dq'/dq = \mu^{-2}$ at the fixed point, the round trip rotates $\mathbb{H}$ about the mode by 2Θ, twice the round-trip Gouy phase of the TEM_{00} mode. The trace classification, the Gouy phase and the stability diagram are one fact.

The Gouy phase is an area. Simon and Mukunda showed that the Gouy phase is a *geometric* phase of this hyperbolic geometry [39]; an accessible derivation is in [40]. In the form we checked

numerically: let $\mathcal{A}(\tau)$ be the signed hyperbolic area of the geodesic triangle with vertices 0 (a point source), τ (the beam) and ∞ (a plane wave). Then the free-space Gouy phase per transverse dimension is $\frac{1}{2}\arctan(z/z_R) = \frac{1}{4}\mathcal{A}(\tau)$. So the Gouy phase accumulated between two planes is a quarter of the hyperbolic area swept out by the beam parameter, measured against these two reference states. Subbarao gives a related Berry-phase reading for general ABCD systems [41].

8 Generalizations

Two transverse dimensions. Ray matrices are now 4×4 elements of $\mathrm{Sp}(4,\mathbb{R})$, which is a stronger condition than $\det = 1$ [7]. A general astigmatic Gaussian $\exp(\frac{ik}{2}\,\mathbf{x}^T\mathbf{q}^{-1}\mathbf{x})$ is annihilated by a two-dimensional (Lagrangian) space of complex rays, the columns of $\binom{\mathbf{q}}{I}$. Theorem 1 goes through word for word:

$$\mathbf{q}' = (\mathbf{A}\mathbf{q}+\mathbf{B})(\mathbf{C}\mathbf{q}+\mathbf{D})^{-1}, \qquad \text{prefactor } \det(\mathbf{A}+\mathbf{B}\mathbf{q}^{-1})^{-1/2} \tag{16}$$

[10, 17, 42]. The space of beams is the Siegel upper half-space $\mathrm{Sp}(4,\mathbb{R})/\mathrm{U}(2)$ [28, 43, 44]. It has rank two, so two astigmatic beams are separated by two invariant distances, not one.

Partially coherent beams. Gaussian Schell-model beams obey a generalized ABCD law through their Wigner functions [7, 17]. The parameter used by Barriuso et al. is of this kind [29].

Lossy elements. Gaussian apertures and gain profiles give complex ABCD matrices [2, 19]. These form a semigroup, not a group: they map $\mathbb{H}$ into itself without being invertible on it. Mathematically this is the oscillator semigroup [44].

9 Summary

optics	mathematics
ABCD matrix, $\det M = 1$	$\mathrm{SL}(2,\mathbb{R}) = \mathrm{Sp}(2,\mathbb{R})$; Liouville
Gaussian beam with parameter q	complex ray $v = (q, 1)$; point of $\mathbb{C}P^1$
ABCD law	linear action on $\mathbb{C}^2$ seen projectively (Möbius)
why it is exact	Heisenberg evolution of $(\hat{x}, \hat{p})$ is the ray map; u_q is annihilated by $\omega(v, \hat{r})$
$\mathrm{Im}\, q$ keeps its sign	conserved Lagrange invariant $\omega(\bar{v}, v)$
magnification, Gouy phase	$x'/x = A + B/q$ (automorphy factor)
sign of the field	metaplectic double cover $\mathrm{Mp}(2,\mathbb{R}) \to \mathrm{SL}(2,\mathbb{R})$
beam matrix, M^2	$\Sigma \propto \mathrm{Re}(vv^\dagger)$; hyperboloid in $\mathbb{R}^{2,1}$, Lorentz-invariant mass
space of Gaussian beams	hyperbolic plane $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$
mode-matching efficiency	$\mathrm{sech}^2(d_{\mathbb{H}}/2)$
stable / marginal / unstable cavity	elliptic / parabolic / hyperbolic isometry
round-trip Gouy phase	eigenvalue of the complex ray; rotation angle about the mode
Gouy phase through focus	$\frac{1}{4}\times$ hyperbolic area
astigmatic beams	Siegel upper half-space $\mathrm{Sp}(4,\mathbb{R})/\mathrm{U}(2)$

A reading path

- *History and practice:* Kogelnik and Li [3], the canonical tutorial; Kogelnik's 1965 BSTJ paper [1], the original law and its name; Siegman ch. 20 [19], the Huygens integral in ABCD form and complex ABCD matrices.
- *Complex rays:* Arnaud 1970 [10], freely available in the BSTJ archive. Its first sections are the clearest physical account of why the law holds.
- *Phase:* Erden and Ozaktas [4], short, with two proofs of the Gouy formula; Ciobanu et al. [25], why the sign matters in practice.
- *Group theory:* Arvind, Dutta, Mukunda and Simon [16], the best single review at this level (free on arXiv); Littlejohn [15] §§2.1, 4, 8 for the wave-packet view and the double cover.
- *Geometry:* Barriuso et al. [29], six pages, beams in the Poincaré half-plane and resonators; Mukunda, Chaturvedi and Simon [40], the Gouy phase as an area; Katok [36] or Needham [37] for the hyperbolic plane itself.

A Numerical checks

Every formula above was checked against direct numerical solution of the paraxial wave equation (FFT angular-spectrum propagation and exact thin-lens phase masks, which do not use the ABCD formalism). The scripts are `scripts/verify_abcd.py` and `scripts/check_gouy_area.py`.

- *ABCD law and Gouy phase.* We used a nine-element system (five free-space sections, four lenses) and HG_n input, $n = 0, \dots, 3$. The numerically propagated field matched $e^{-i(n+\frac{1}{2})\psi}\,\mathrm{HG}_n(q')$ with overlap $\geq 1-10^{-7}$ and phase error $\leq 2\times10^{-9}$ rad. The accumulated $\psi = 5.9207$ differs from the principal value $\arg(A + B/q_0) = -0.3625$ by 2π.
- *Overlap.* For random pairs of beams, $|\langle u_1|u_2\rangle|^2$ agreed with $\mathrm{sech}(d/2)$ to 10^{-10}.
- *Moments.* Equation (13) held to 10^{-5}, and $\det\Sigma/(\lambda/2)^2 = 1.000001$.
- *Resonator.* For a two-mirror cavity, $\arccos\frac{A+D}{2} = \arg(Cq_* + D) = 2\arccos\sqrt{g_1 g_2} = 2.117647$. The numerical round-trip phase of the eigenmode was $-1.058824 = -\Theta/2$.
- *Metaplectic sign.* One full period of a 1D graded-index guide ($M = I$) returned $\langle u_0|u\rangle = -1.000000$.
- *Gouy area.* $\frac{1}{4}\mathcal{A}(\tau_2) - \frac{1}{4}\mathcal{A}(\tau_1) = \frac{1}{2}[\arctan(z_2/z_R) - \arctan(z_1/z_R)]$ to 10^{-6} for waists on either side.

References

[1] H. Kogelnik. Imaging of optical modes—resonators with internal lenses. *Bell System Technical Journal*, 44(3):455–494, 1965. doi: 10.1002/j.1538-7305.1965.tb01672.x.

[2] H. Kogelnik. On the propagation of Gaussian beams of light through lenslike media including those with a loss or gain variation. *Applied Optics*, 4(12):1562–1569, 1965. doi: 10.1364/AO.4.001562.

[3] H. Kogelnik and T. Li. Laser beams and resonators. *Applied Optics*, 5(10):1550–1567, 1966. doi: 10.1364/AO.5.001550.

[4] M. F. Erden and H. M. Ozaktas. Accumulated Gouy phase shift in Gaussian beam propagation through first-order optical systems. *J. Opt. Soc. Am. A*, 14(9):2190–2194, 1997. doi: 10.1364/JOSAA.14.002190.

[5] V. Guillemin and S. Sternberg. *Symplectic Techniques in Physics*. Cambridge University Press, Cambridge, 1984.

[6] K. B. Wolf. *Geometric Optics on Phase Space*. Springer, Berlin, 2004.

[7] R. Simon, E. C. G. Sudarshan, and N. Mukunda. Generalized rays in first-order optics: transformation properties of Gaussian Schell-model fields. *Phys. Rev. A*, 29(6):3273–3279, 1984. doi: 10.1103/PhysRevA.29.3273.

[8] G. A. Deschamps. Gaussian beam as a bundle of complex rays. *Electronics Letters*, 7(23): 684–685, 1971. doi: 10.1049/el:19710467.

[9] J. B. Keller and W. Streifer. Complex rays with an application to Gaussian beams. *J. Opt. Soc. Am.*, 61(1):40, 1971. doi: 10.1364/JOSA.61.000040.

[10] J. A. Arnaud. Nonorthogonal optical waveguides and resonators. *Bell System Technical Journal*, 49(9):2311–2348, 1970. doi: 10.1002/j.1538-7305.1970.tb02528.x.

[11] J. A. Arnaud. Hamiltonian theory of beam mode propagation. In E. Wolf, editor, *Progress in Optics*, volume XI, pages 247–304. North-Holland, 1973. doi: 10.1016/S0079-6638(08) 70138-0.

[12] D. Gloge and D. Marcuse. Formal quantum theory of light rays. *J. Opt. Soc. Am.*, 59(12): 1629, 1969. doi: 10.1364/JOSA.59.001629.

[13] D. Stoler. Operator methods in physical optics. *J. Opt. Soc. Am.*, 71(3):334, 1981. doi: 10.1364/JOSA.71.000334.

[14] M. Nazarathy and J. Shamir. First-order optics—a canonical operator representation: lossless systems. *J. Opt. Soc. Am.*, 72(3):356, 1982. doi: 10.1364/JOSA.72.000356.

[15] R. G. Littlejohn. The semiclassical evolution of wave packets. *Physics Reports*, 138(4–5): 193–291, 1986. doi: 10.1016/0370-1573(86)90103-1.

[16] Arvind, B. Dutta, N. Mukunda, and R. Simon. The real symplectic groups in quantum mechanics and optics. *Pramana – J. Phys.*, 45(6):471–497, 1995. doi: 10.1007/BF02848172. arXiv:quant-ph/9509002.

[17] R. Simon, E. C. G. Sudarshan, and N. Mukunda. Gaussian pure states in quantum mechanics and the symplectic group. *Phys. Rev. A*, 37(8):3028–3038, 1988. doi: 10.1103/PhysRevA.37. 3028.

[18] Jr. S. A. Collins. Lens-system diffraction integral written in terms of matrix optics. *J. Opt. Soc. Am.*, 60(9):1168–1177, 1970. doi: 10.1364/JOSA.60.001168.

[19] A. E. Siegman. *Lasers*. University Science Books, Mill Valley, CA, 1986. Ch. 20 ("Huygens' integral and ABCD matrices", "Gaussian beams and ABCD matrices", complex ABCD matrices).

[20] E. J. Heller. Time-dependent approach to semiclassical dynamics. *J. Chem. Phys.*, 62(4): 1544–1555, 1975. doi: 10.1063/1.430620.

[21] G. A. Hagedorn. Semiclassical quantum mechanics. I. *Commun. Math. Phys.*, 71(1):77–93, 1980. doi: 10.1007/BF01230088.

[22] T. Ohsawa. The Siegel upper half space is a Marsden–Weinstein quotient: symplectic reduction and Gaussian wave packets. *Lett. Math. Phys.*, 105(9):1301–1320, 2015. doi: 10.1007/s11005-015-0780-z. arXiv:1504.03963.

[23] H. Bacry and M. Cadilhac. Metaplectic group and Fourier optics. *Phys. Rev. A*, 23(5): 2533–2536, 1981. doi: 10.1103/PhysRevA.23.2533.

[24] R. Simon and K. B. Wolf. Structure of the set of paraxial optical systems. *J. Opt. Soc. Am. A*, 17(2):342, 2000. doi: 10.1364/JOSAA.17.000342.

[25] A. A. Ciobanu, D. D. Brown, P. J. Veitch, and D. J. Ottaway. Modeling circulating cavity fields using the discrete linear canonical transform. *J. Opt. Soc. Am. A*, 38(9):1293, 2021. doi: 10.1364/JOSAA.433575. arXiv:2106.03367.

[26] M. J. Bastiaans. Wigner distribution function and its application to first-order optics. *J. Opt. Soc. Am.*, 69(12):1710, 1979. doi: 10.1364/JOSA.69.001710.

[27] P. A. Bélanger. Beam propagation and the ABCD ray matrices. *Optics Letters*, 16(4): 196–198, 1991. doi: 10.1364/OL.16.000196.

[28] T. Ohsawa and C. Tronci. Geometry and dynamics of Gaussian wave packets and their Wigner transforms. *J. Math. Phys.*, 58:092105, 2017. doi: 10.1063/1.4995233. arXiv:1611.04081.

[29] A. G. Barriuso, J. J. Monzón, L. L. Sánchez-Soto, and J. F. Cariñena. Geometrical aspects of first-order optical systems. *J. Opt. A: Pure Appl. Opt.*, 7(9):451–456, 2005. doi: 10.1088/1464-4258/7/9/002. arXiv:physics/0506112.

[30] L. L. Sánchez-Soto, J. J. Monzón, A. G. Barriuso, and J. F. Cariñena. The transfer matrix: a geometrical perspective. *Physics Reports*, 513(4):191–227, 2012. doi: 10.1016/j.physrep.2011.10.002. arXiv:1205.1318.

[31] R. Simon and N. Mukunda. Iwasawa decomposition in first-order optics: universal treatment of shape-invariant propagation for coherent and partially coherent beams. *J. Opt. Soc. Am. A*, 15(8):2146–2155, 1998. doi: 10.1364/JOSAA.15.002146.

[32] F. Bayer-Helms. Coupling coefficients of an incident wave and the modes of a spherical optical resonator in the case of mismatching and misalignment. *Applied Optics*, 23(9):1369, 1984. doi: 10.1364/AO.23.001369.

[33] A. A. Ciobanu, D. D. Brown, P. J. Veitch, and D. J. Ottaway. Mode matching error signals using radio-frequency beam shape modulation. *Applied Optics*, 59(31):9884, 2020. doi: 10.1364/AO.404646. arXiv:2008.05839.

[34] E. M. Rabei, Arvind, N. Mukunda, and R. Simon. Bargmann invariants and geometric phases: a generalized connection. *Phys. Rev. A*, 60(5):3397–3409, 1999. doi: 10.1103/PhysRevA.60.3397. arXiv:quant-ph/9907080.

[35] A. Perelomov. *Generalized Coherent States and Their Applications*. Springer, Berlin, 1986.

[36] S. Katok. *Fuchsian Groups*. Chicago Lectures in Mathematics. University of Chicago Press, Chicago, 1992.

[37] T. Needham. *Visual Complex Analysis*. Oxford University Press, Oxford, 1997.

[38] S. Başkal and Y. S. Kim. ABCD matrices as similarity transformations of Wigner matrices and periodic systems in optics. *J. Opt. Soc. Am. A*, 26(9):2049–2054, 2009. doi: 10.1364/JOSAA.26.002049. arXiv:0805.2604.

[39] R. Simon and N. Mukunda. Bargmann invariant and the geometry of the Güoy effect. *Phys. Rev. Lett.*, 70(7):880–883, 1993. doi: 10.1103/PhysRevLett.70.880.

[40] N. Mukunda, S. Chaturvedi, and R. Simon. Classical light beams and geometric phases. *J. Opt. Soc. Am. A*, 31(6):1141, 2014. doi: 10.1364/JOSAA.31.001141. arXiv:1312.3987.

[41] D. Subbarao. Topological phase in Gaussian beam optics. *Optics Letters*, 20(21):2162–2164, 1995. doi: 10.1364/OL.20.002162.

[42] J. A. Arnaud and H. Kogelnik. Gaussian light beams with general astigmatism. *Applied Optics*, 8(8):1687–1693, 1969. doi: 10.1364/AO.8.001687.

[43] C. L. Siegel. Symplectic geometry. *American Journal of Mathematics*, 65(1):1–86, 1943. doi: 10.2307/2371774.

[44] G. B. Folland. *Harmonic Analysis in Phase Space*, volume 122 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, 1989.